In-class Template

Natalie Chen

1. Set up

Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# read in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```r
# reads from taxon list from the data file
taxon_list <- read_csv(here("data","taxon_list.csv"))

# aquatic invertebrates
```

```r
aq_ins <- read_xlsx(here("data","Aquatic Sampling Data-2026-03-10.xlsx"),
  ↪  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data","Aquatic Sampling Data-2026-03-10.xlsx"),
  ↪  sheet = "sites")

# water quality

water_quality <-read_xlsx(
  here("data","Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",

  # set any string in this character vector to NA.
  na = c("","n/a","N/A","over","173+"))
```

2. Cleaning

`sites` object

Goal: Create a new object from 'sites" that only includes NCOS sites

```r
# creating new object from sites
ncos_sites <- sites |>

  # filter to only include NCOS quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

clean `taxon_list`

Goal: convert character strings in the existing `taxon_list` data frame into snakecase

```r
# creating new object from taxon_list
taxon_list_clean <- taxon_list |>

  # converted taxon name values to snake case (lowercase)
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

clean `water_quality`

Goal: create a data frame with median pH, median DO (mg/L), median salinity (ppt) for each date and site

pseudocode

- create new object from `water_quality`
- clean column names using janitor
- only include surveys from quarterly NCOS sites
- create new column called `date_site` to join w later data frames
- group by date_site
- calculate median pH, salinity, and dissolved oxygen
- ungroup
- filter out any outliers

```r
water_quality_clean <- water_quality |>

  # cleaning column names
  clean_names() |>

  # filtering join: only include sites in ncos_sites
  # keep all the rows on the left that match up with rows on the right
  semi_join(ncos_sites, by = "site") |>

  # combine date and site together and remove those indiv. columns
  unite("date_site", date, site, remove = TRUE) |>

  # group by date_site
  group_by(date_site) |>

  # create new columns that show median of each variable at each date_site
  # because there are multiple samples at the same site at different depths
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>

  # ungroup
  ungroup() |>

  # filter out the salinity outlier
  filter(date_site != "2022-08-12_NVBR")
```

clean `aq_ins`

Goal: create a long format data frame in which each row contains the counted individuals in each taxon during each survey. The data frame should include taxonomic info for each organism and the water quality info for each survey

- create a new object from `aq_ins`
- clean column names
- filter to only include quarterly NCOS sites
- select columns
- filter to only include "filtered beaker" sample
- convert the data frame to long format
- group by site, date, and taxon
- calculate average abundance per liter (sum of counts then divided by 7.5 liters)
- create a new column called `data_site` to join with water quality
- join data frames
- join w taxon list

```
# creating new clean aquatic inverts object from aquatic inverts
aq_ins_clean <- aq_ins |>

  # cleaning column names
  clean_names() |>

  # filtering join: only include quarterly NCOS surveys from ncos_sites
  semi_join(ncos_sites, by = "site") |>

  # renaming column to just date
  rename(date = date_on_vial) |>

  # select columns of interest
  select(date, sample_type, site,

         # 9 most abundant taxon
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
```

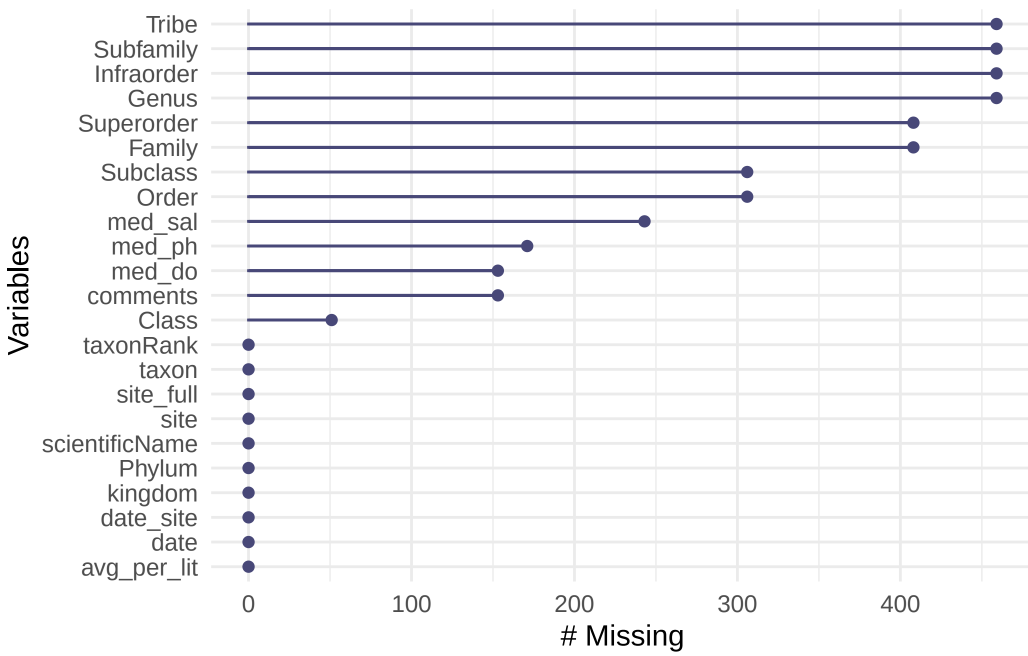
```r
# filter to only include filtered beaker samples
filter(sample_type %in% c("FB250", "FB 250"))  |>


pivot_longer(
  # cols = columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
) |>

# group by the date, site, and taxon
group_by(date, site, taxon) |>

# calculate average abundance per liter (sum abundance divided by 7.5
 ↪  liters)
summarize(avg_per_lit = sum(abundance, na.rm = TRUE)/7.5) |>

# ungroup
ungroup() |>

 # create a new column called `date_site` to join with water quality
 # don't remove the original columns
unite("date_site", date, site, remove = FALSE) |>

# join with water quality clean
# wqc has `date_site` column as well
left_join(water_quality_clean, by = "date_site") |>

# join with taxonomic information from taxon list
# verbatim_name column in taxon_list_clean matches with taxon column in
 ↪  aq_ins_clean
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

# fill in full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "East Channel",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
```

```
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>

  # setting factor levels for full site name by median salinity
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
   ↪  = TRUE),
         site_full = fct_rev(site_full))
```

3. Visualizing

Visualize missing values

```
gg_miss_var(aq_ins_clean)
```

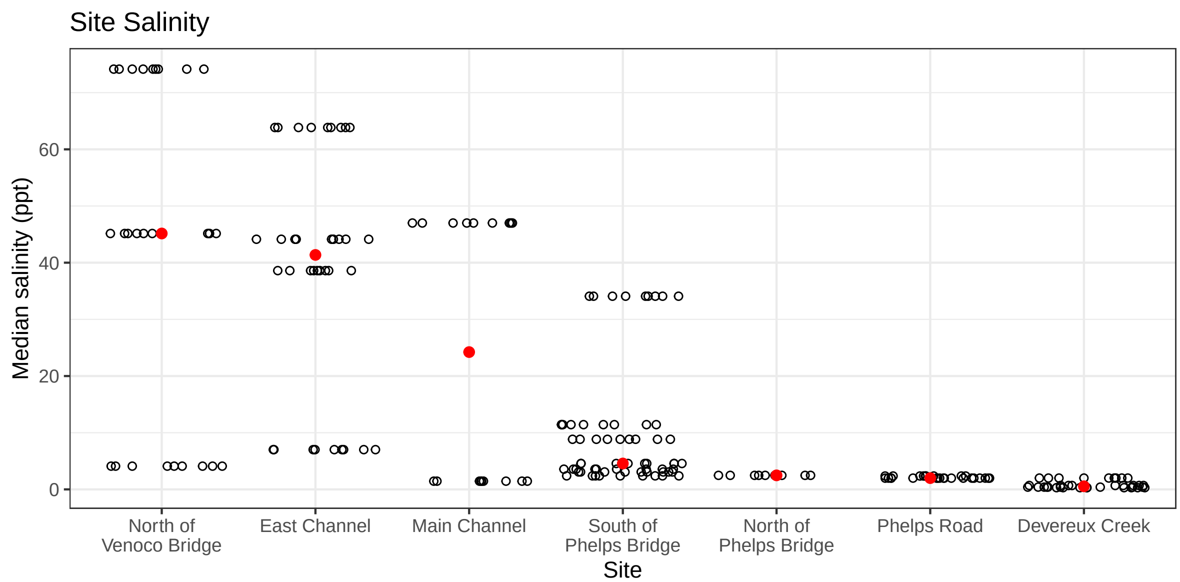


Missing values for salinity, pH, and dissolved oxygen might influence later figures.

Visualize salinity

- x-axis: `site`
- y-axis: `mean_sal`
- geometry to show salinity during survey: `geom_jitter`
- geometry to show median salinity: `stat_summary`

```
salinity_plot <- ggplot(data = aq_ins_clean,
                        # x-axis: full site name
                        # y-axis: median salinity
                        mapping = aes(x = site_full,
                                      y = med_sal)) +
  # points representing median salinity at each survey
  geom_jitter(height = 0,
              shape = 21) +

  stat_summary(geom = "point",
               fun = median,
               color = "red",
               size = 2,) +

  # wrap x-axis labels
  scale_x_discrete(labels = label_wrap(width = 15)) +

  # labeling titles and axes
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site Salinity") +

  # change theme
  theme_bw()


# displaying the object
salinity_plot
```



Sites seem to differ in salinity. North of Venoco Bridge, East Channel, and Main Channel tends to be hypersaline, while North of Phelps Road and Devereux Creek is the least saline. South of Phelps Bridge seems to be more freshwater but is higher in salinity than the other, lower salinity sites.

Visualization differences in copepod abundance

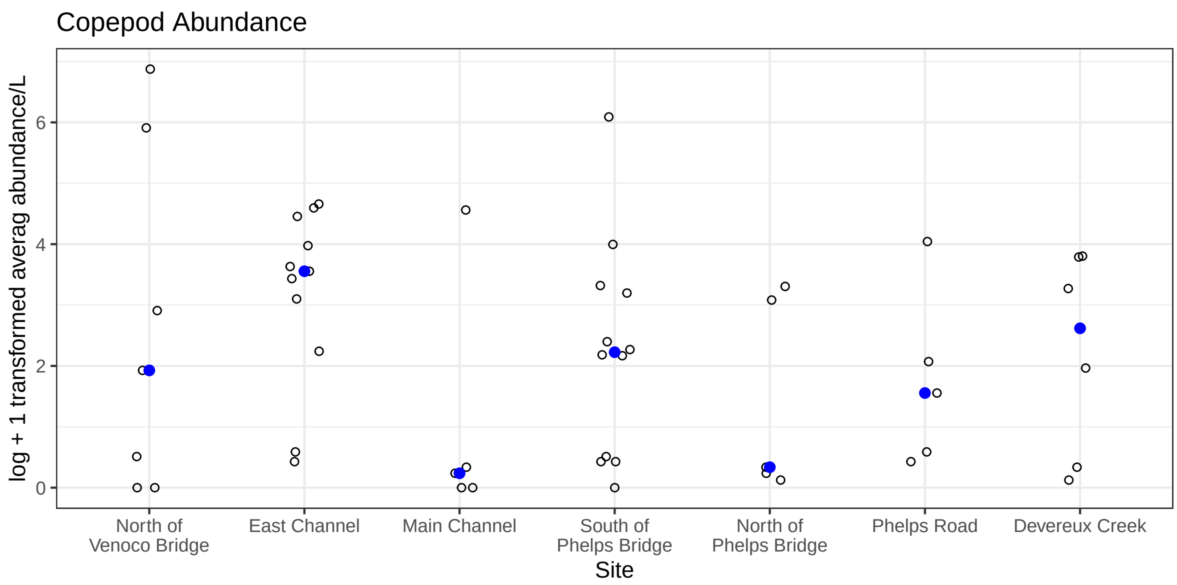
- x-axis: site_full
- y-axis: average abundance per liter
- geometry to represent observations: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

Initial cleaning:

```
copepods <- aq_ins_clean |>

  filter(taxon == "copepod") |>

  mutate(log_avg_per_lit = log(avg_per_lit + 1))
```

Visualizing:

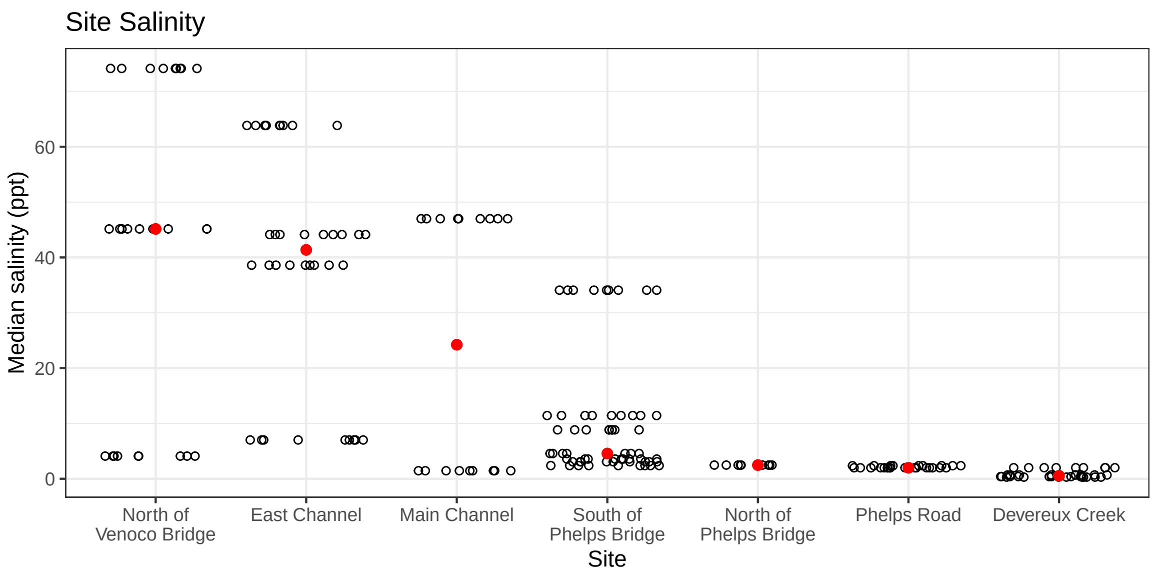
```
# creating object to store plot
copepod_plot <- ggplot(data = copepods,
                       # x-axis: full site name
                       # y-axis: log transformed avg count/L
                       mapping = aes(x = site_full,
                                     y = log_avg_per_lit)) +
  # jitter to represent salinity
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
# points to represent median copepod abundance
  stat_summary(geom = "point",
               fun = median,
               color = "blue",
               size = 2) +
  # wrapping x axis labels
scale_x_discrete(labels = label_wrap(width = 15)) +

  # relabelling axes
  labs(x = "Site",
       y = "log + 1 transformed averag abundance/L",
       title = "Copepod Abundance") +

  # different theme
  theme_bw()

# display object
copepod_plot
```

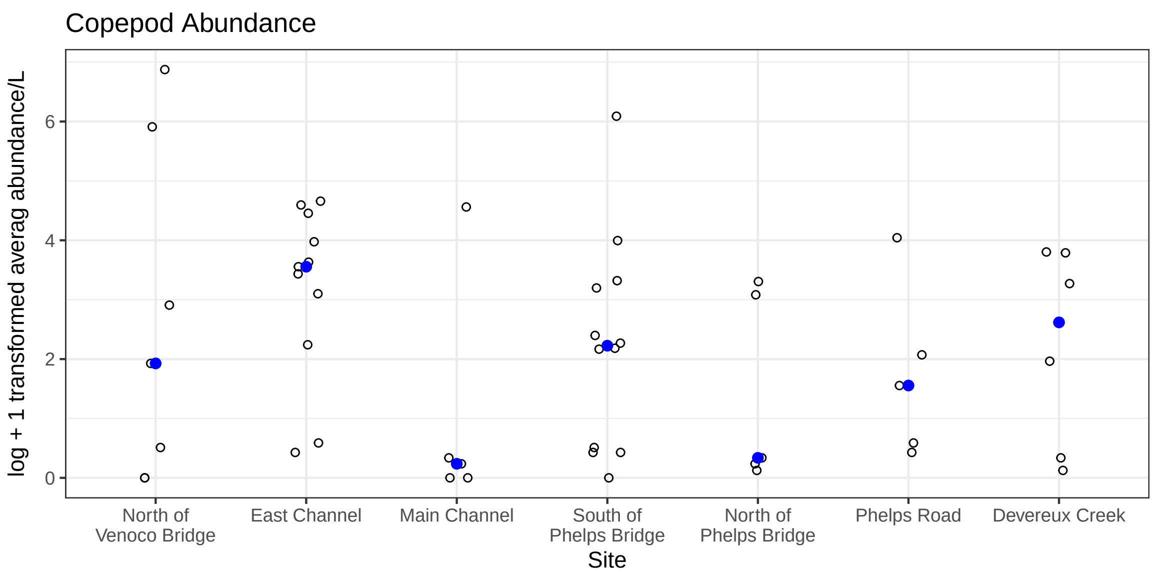


Copepod abundance varies across sites.

Putting salinity and copepod abundance together

```
salinity_plot / copepod_plot
```


Site Salinity
Median salinity (ppt)
0
20
40
60
North of
Venoco Bridge
East Channel
Main Channel
South of
Phelps Bridge
North of
Phelps Bridge
Phelps Road
Devereux Creek
Site


Copepod Abundance
log + 1 transformed averag abundance/L
0
2
4
6
North of
Venoco Bridge
East Channel
Main Channel
South of
Phelps Bridge
North of
Phelps Bridge
Phelps Road
Devereux Creek
Site